

MAT282 · INTERMEDIATE STATISTICS · FALL 2026 · VERSION A

Answer key

Version A

Marking guide and worked solutions · 40 points · instructor copy: do not distribute before the exam

How to mark. Sections 1 and 5 and the explanation parts of the others reward **sensible reasoning that is defended with a reason or a number**; there is no single right model. Apply **follow-through** from earlier answers. Accept rounding differences of ± 0.02 in s and t and ± 0.5 in forecasts. Rules used in this paper: a coefficient is **clearly different from zero if it is more than 2 standard errors from zero**; a t -test is **significant if $p < 0.05$** (two-tailed, T.TEST type 2); critical t for $df = 8$ is 2.31. Objective column: syllabus learning outcome (LO) and your category, **Con** concept, **Cal** calculation, **Com** computation in Sheets.

1 · Quick answer sheet

Q	Objective	Expected answer	Pts	How the marks split
1a	LO1, LO5 · Con	Any three sensible inputs with a direction and a reason: temperature (up: hot days want cold treats), weekend (up: more leisure time), beach event (up: crowds nearby), planned price (down: dearer gelato sells less), sunshine (up, but overlaps with temperature).	3	1 per input: direction and a coherent reason. A “wrong” direction with a good reason scores; a direction with no reason does not.
1b	LO5 · Con	Item 6 , iced coffee sales counted at closing time: it is not known before opening (only after the day) and it is driven by the same weather and day type, so it cannot be used to forecast.	2	1 identifies item 6 + 1 the problem (not known in advance).
1c	LO3, LO5, LO6 · Con	Not both. With $r = 0.78$ they carry mostly the same information, so LINEST cannot separate their effects: coefficients become unstable and their standard errors large (the model assumes inputs that are not duplicates of each other). Keep temperature ($r = 0.58$ with sales vs 0.49; a more direct measure of heat).	2	1 overlap and why it is a problem + 1 a choice with a reason.
1d	LO6 · Con	Hold back some days, fit the model on the rest, predict the held-back days, then calculate their residuals and MAE.	1	1: a hold-out check with residuals/MAE (names the idea of unseen days).
2a	LO5, LO6 · Com	Sales = $72.87 + 4.31 \times \text{temperature} + 1.99 \times \text{sunshine} + 27.09 \times \text{event}$ ($\hat{y} = b + m_1x_1 + m_2x_2 + m_3x_3$; the table shows Col A = m_3 , Col B = m_2 , Col C = m_1 , Col D = b).	2	1 correct numbers matched to the right variables + 1 correct equation form.
2b	LO6, LO7 · Con	(i) 72.87 is the predicted sales when temperature = 0°C , sunshine = 0 h and no event; it is not the sales on an average day (and 0°C is outside the data). (ii) On event days the model predicts about 27 more cups than on days without an event, holding temperature and sunshine fixed; an association, not proof that events cause the sales. (iii) About 43% of the day-to-day variation in sales is explained; a typical forecast misses by about 27 cups ($SE - Y$).	3	1 each: the corrected statement must keep units and “holding the others fixed”.
2c	LO6 · Cal + Con	(a) Sunshine : $ 1.99 $ is less than $2 \times 1.37 = 2.74$, so it is not clearly different from zero; temperature (4.31 vs $2 \times 1.01 = 2.02$) and event (27.09 vs 10.36) are. (b) Sunshine overlaps with temperature ($r = 0.78$); once temperature is in the model it adds almost nothing new. (c) Drop sunshine and refit with temperature and event.	3	1 correct predictor with the comparison + 1 overlap reason + 1 sensible action.
2d	LO5, LO6 · Cal	(a) $\hat{y} = 72.87 + 4.31(26) + 1.99(8) + 27.09(1) = \mathbf{227.9}$ cups. (b) $2 \times 27.22 = 54.44$, so about 174 to 282 cups. (c) 150 is outside the range, so it is surprising: more than $2 SE - Y$ from the prediction (a storm, a closure, a stock-out or a special event?); investigate.	4	2 prediction (1 substitution, 1 value) + 1 range + 1 judgement with a reason.
3a-i	LO3, LO6 · Cal	Filled table: Mon actual 180, predicted 190, residual -10 ; Sat actual 245, predicted 230, residual $+15$; Wed actual 160, predicted 165, residual -5 .	1	1 all blanks correct (follow-through on a single slip).
3a-ii	LO3, LO6 · Cal	MAE = $(10 + 15 + 5) \div 3 = \mathbf{10}$ cups.	1	1 correct (absolute values used).
3a-iii	LO6 · Con	Model Z has the lower MAE (6 vs 10). But every residual has the same sign: Z over-predicts every day by 6 (systematic bias). A simple correction (subtract 6) would remove it. Model X’s errors are larger but not one-sided.	2	1 lower MAE with numbers + 1 identical signs = bias.

3b	LO3, LO4, LO6 · Con	(i) Day 8 : the hottest day, but sales far below the pattern. (ii) Without it the slope is 6.29 vs 2.49 with it, R^2 0.97 vs 0.17 (SE – Y 5.2 vs 28.8): one day cuts the slope by more than half and wrecks the fit. (iii) Find out why (closure, power cut, stock-out, recording error). Delete only with a documented reason; otherwise keep it and report results both ways.	4	1 day + 2 impact (1 slope, 1 R^2) + 1 sensible next step.
4a	LO3 · Cal	Means 200 and 186; $\Sigma(x - \bar{x})^2 = 320$ and 256; $s_1 = \sqrt{(320 \div 4)} = 8.94$, $s_2 = \sqrt{(256 \div 4)} = 8.00$ (deviations in the worked table below).	3	1 means + 1 sums of squares + 1 s values.
4b	LO3 · Cal	$s_p = \sqrt{[(4 \times 80 + 4 \times 64) \div 8]} = \sqrt{72} = 8.49$; $t = (200 - 186) \div [8.49 \times \sqrt{(1/5 + 1/5)}] = 14 \div 5.37 = \mathbf{2.61}$.	2	1 s_p + 1 t (follow-through on s; sign may be reported either way if direction is stated).
4c	LO1, LO3, LO6 · Con	(i) H_0 : the price rise made no difference to mean daily gelato sales (mean before = mean after). (ii) $p = 0.031 < 0.05$, so the difference is statistically significant (the same decision as $ t = 2.61 > 2.31$): reject H_0 . (iii) If the price had no effect, a difference this large or larger would turn up in only about 3% of samples like these. It is not the probability that H_0 is true and it does not give the size of the effect.	3	1 null hypothesis + 1 decision with p (and t vs critical value) + 1 correct plain-words meaning.
4d	LO6 · Con	Any sensible other change: temperature or season, a local event, tourist numbers, a menu or staff change; only 5 days per group; this shows association rather than proof of cause.	1	1 a relevant reason/idea.
5	LO7 · Con	(i) The inputs recommended (e.g. temperature and beach events; sunshine adds nothing once temperature is in the model) with the typical forecast error (about 27 cups). (ii) The price test: gelato sales were lower after the price rise (about 14 cups, $p = 0.031$). (iii) A caution (small samples, other changes, association not proof, forecasts can still miss by the typical error).	3	1 inputs + main LINEST finding in plain words + 1 price test result in plain words + 1 a genuine caution.

2 · Worked tables

Section 4 worksheet (Table 6).

Day	Before (x_1)	$x_1 - \bar{x}_1$	$(x_1 - \bar{x}_1)^2$	After (x_2)	$x_2 - \bar{x}_2$	$(x_2 - \bar{x}_2)^2$
1	196	-4	16	184	-2	4
2	204	+4	16	192	+6	36
3	188	-12	144	176	-10	100
4	212	+12	144	196	+10	100
5	200	0	0	182	-4	16
Sum	1000	0	320	930	0	256
Mean $\bar{x}$	200			186		
s	8.94			8.00		

Check against Sheets: =T.TEST gives $p = 0.0312$ ($df = 8$, $|t| = 2.609$, two-tailed critical value 2.306).

3 · Where students go wrong

- **Reading the LINEST table backwards** (2a): Col A is the **last** X column and Col D the intercept. The layout table is printed on the paper, so give no credit for guessing the order; apply follow-through if the order is wrong but the method is right.
- **“Per customer” and “causes”** (2b): the coefficients describe the **daily total** and an **association**.
- **Reading R^2 as “percent correct”** (2b): R^2 is the share of variation explained.
- **Using absolute values** (3a-ii): MAE averages the sizes of the errors, so signs must not cancel.
- **Reading p as the probability that H_0 is true** (4c). Credit “if there were no real effect, a result this extreme would be rare”.